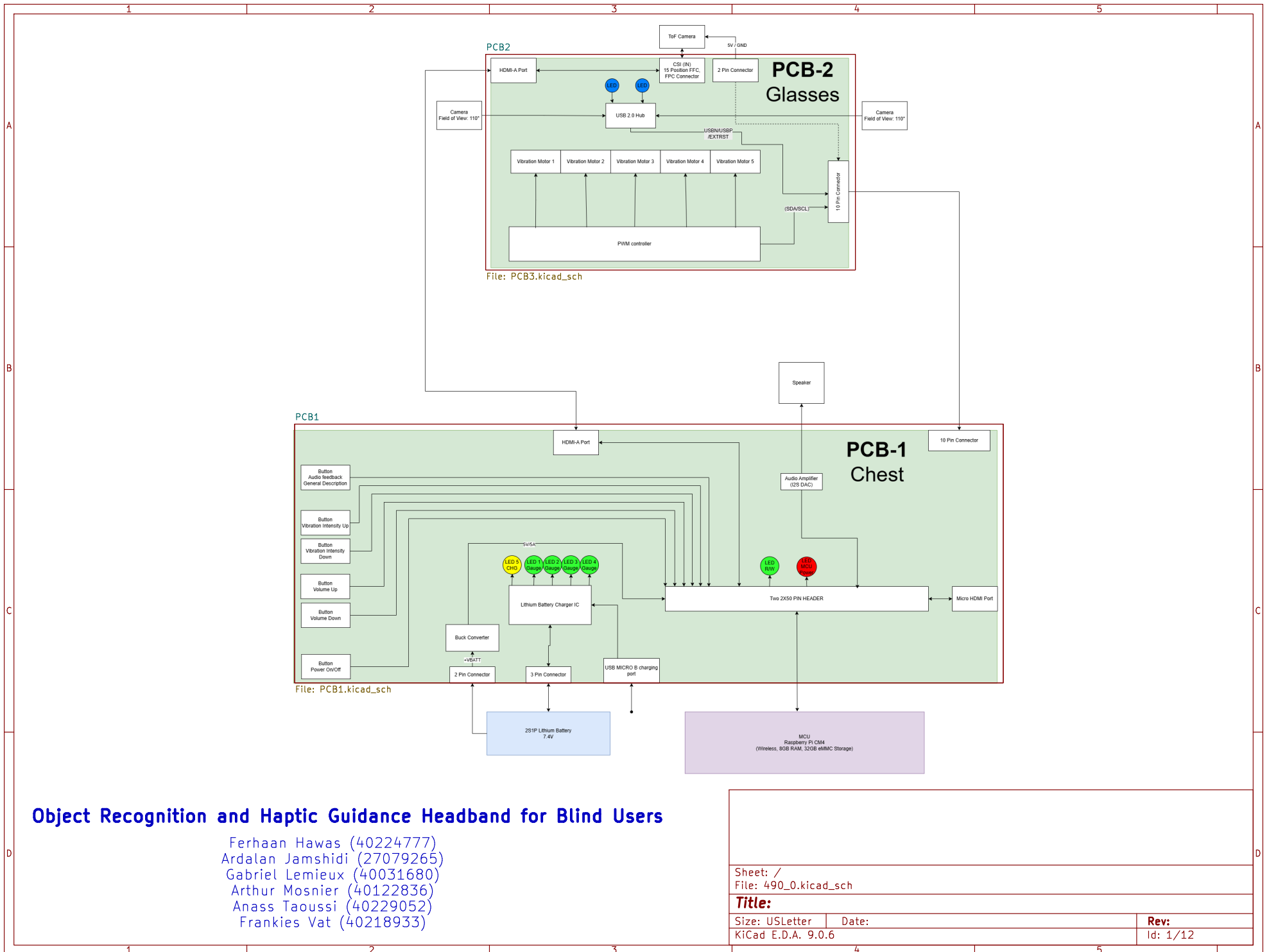




Object Recognition and Haptic Guidance Headband for Blind Users

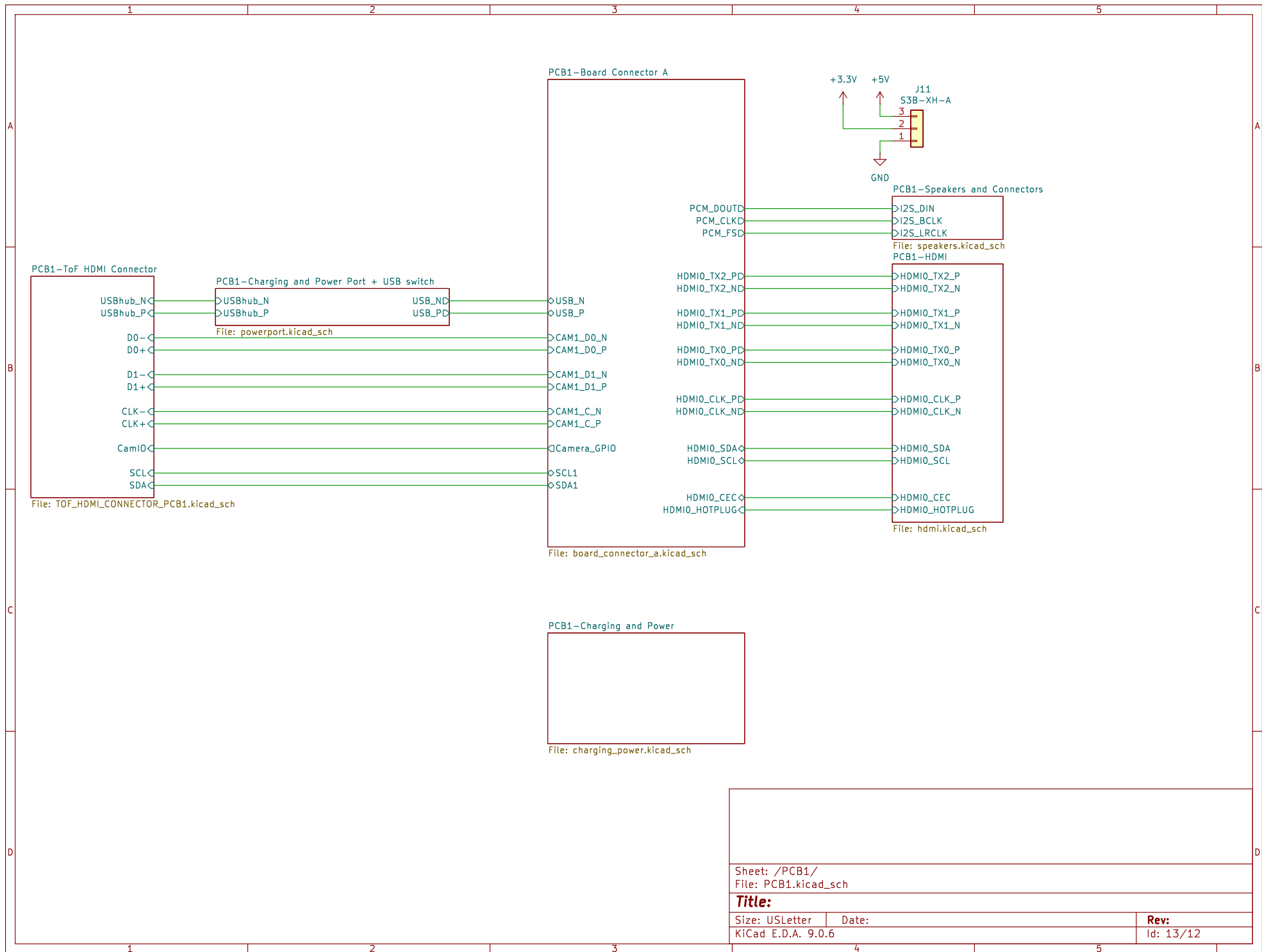
Ferhaan Hawas (40224777)
 Ardalan Jamshidi (27079265)
 Gabriel Lemieux (40031680)
 Arthur Mosnier (40122836)
 Anass Taoussi (40229052)
 Frankies Vat (40218933)

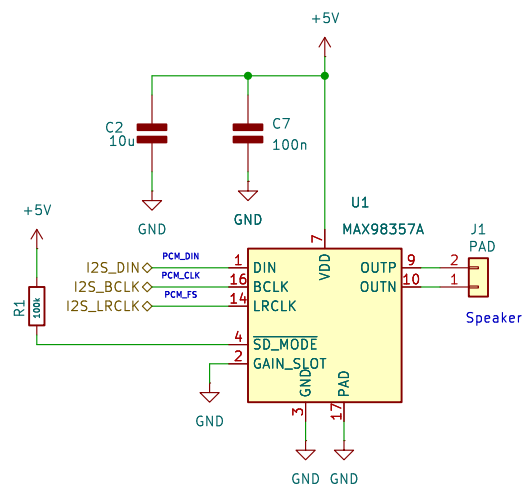
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 File: 490_0.kicad_sch

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Rev:
 Id: 1/12





Sheet: /PCB1/PCB1-Speakers and Connectors/
File: speakers.kicad_sch

Title:

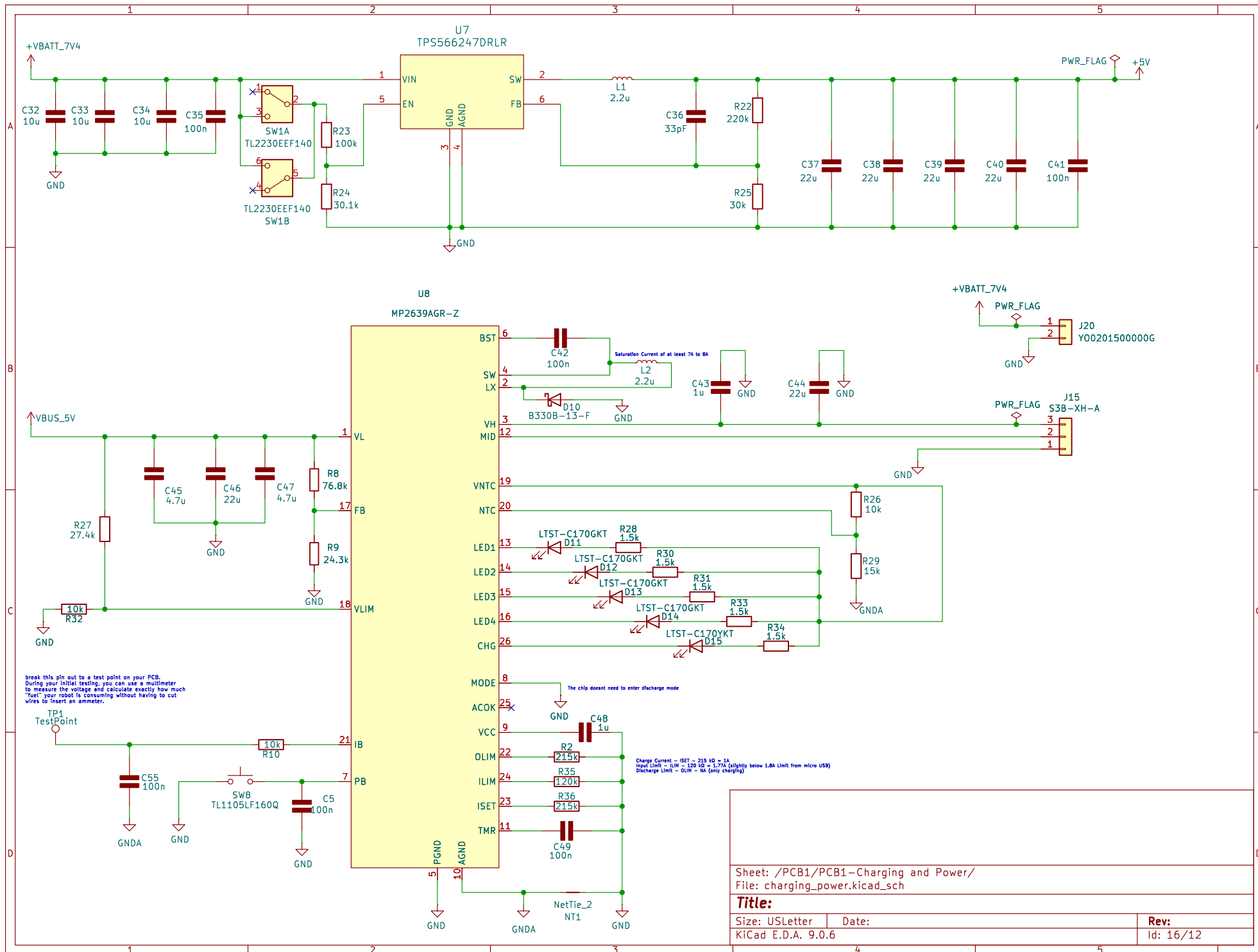
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Id: 2/12



disable the Ethernet controller in software. Even if the pins are disconnected, the internal hardware is still powered by default.

In /boot/config.txt (or /boot/firmware/config.txt), add:

```
Plaintext
dtparam=eth0=off
```

For Flashing (Device Mode): The CM4 ignores this pin when nRPIBOOT is active, but keeping it floating is the "safe default" for device mode.
For Cameras (Host Mode): force the CM4 to be a Host using software. Add this line to config.txt file after flashing the OS:
dtoverlay=dwc2,dr_mode=host

A

B

C

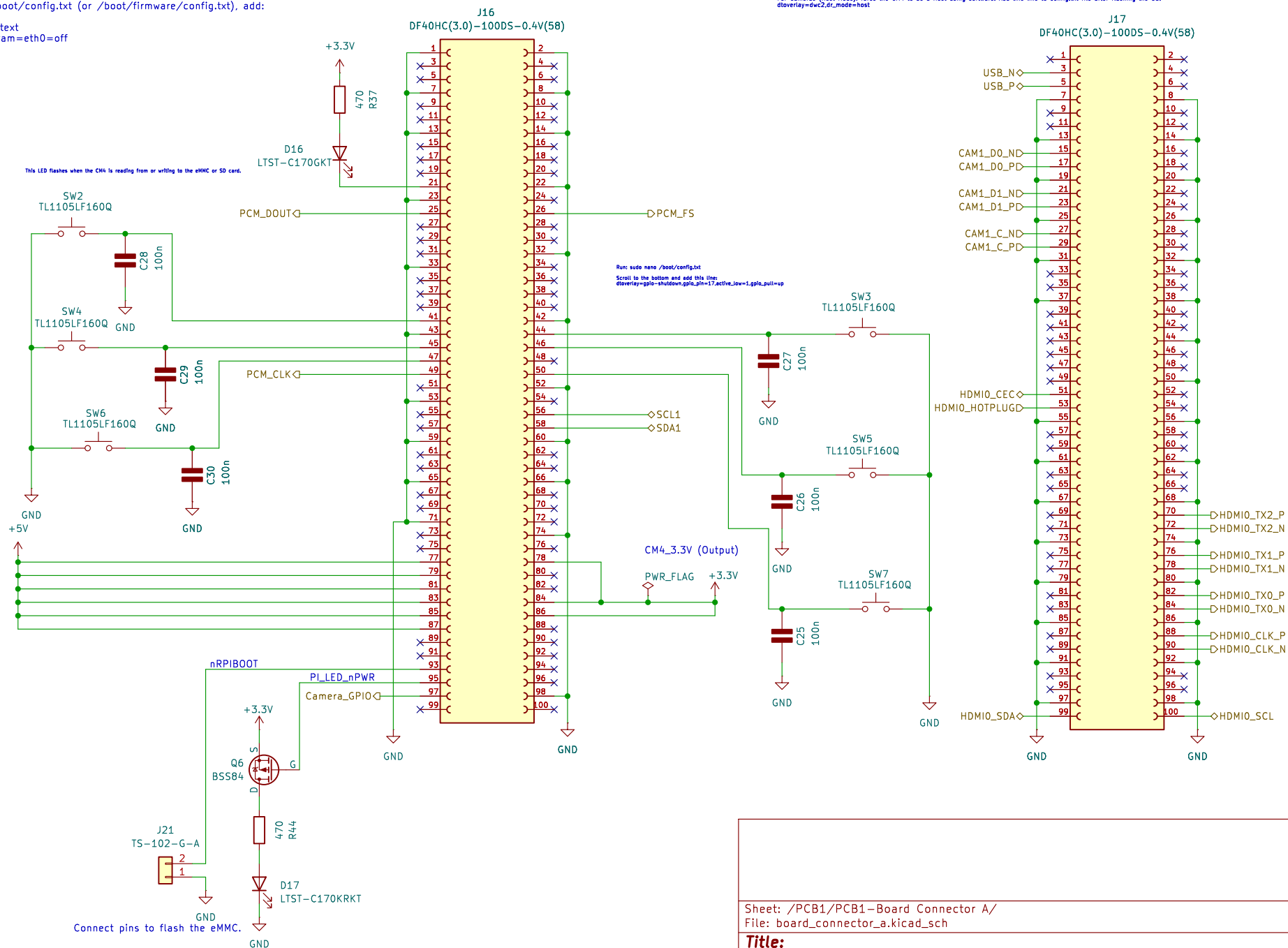
D

A

B

C

D

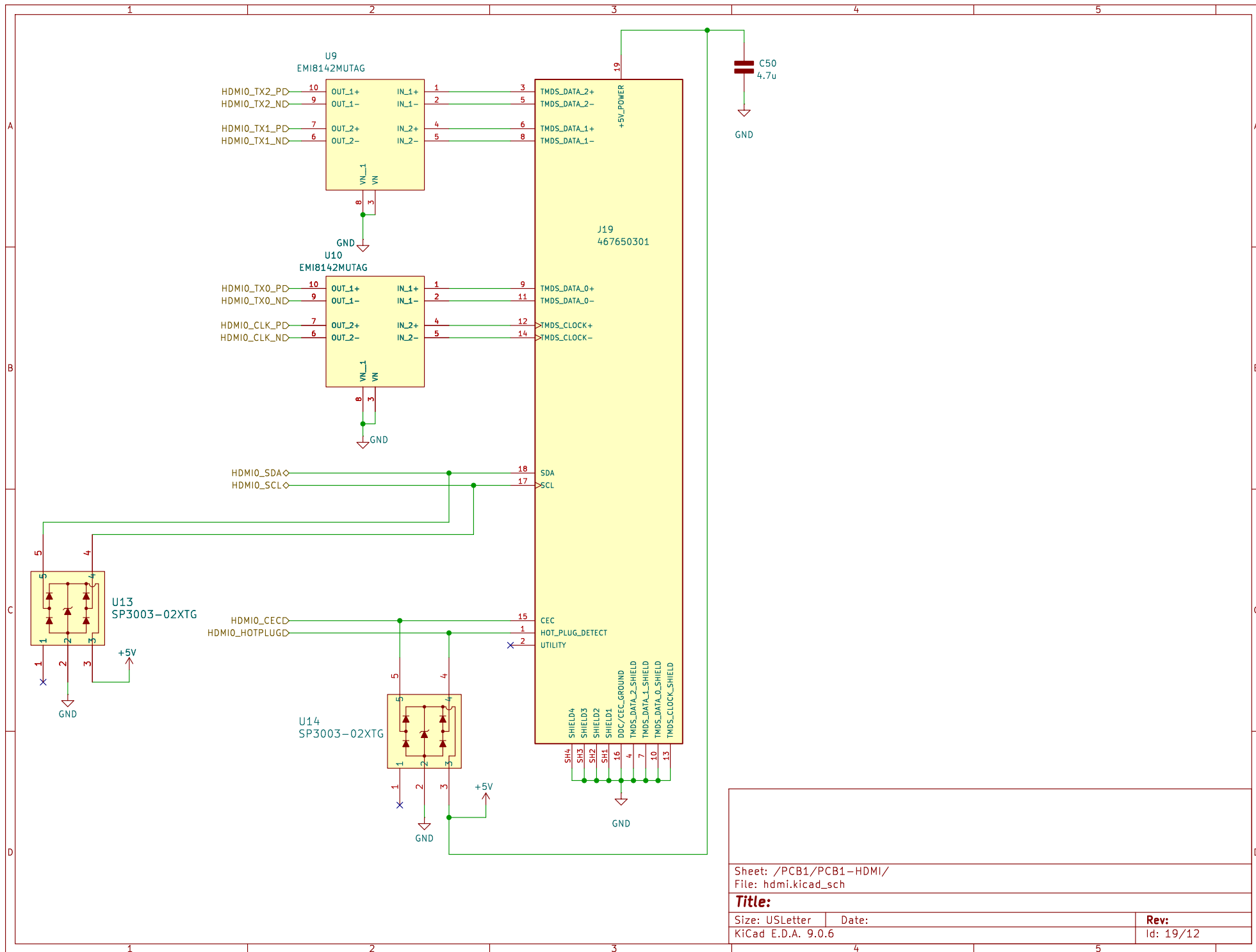


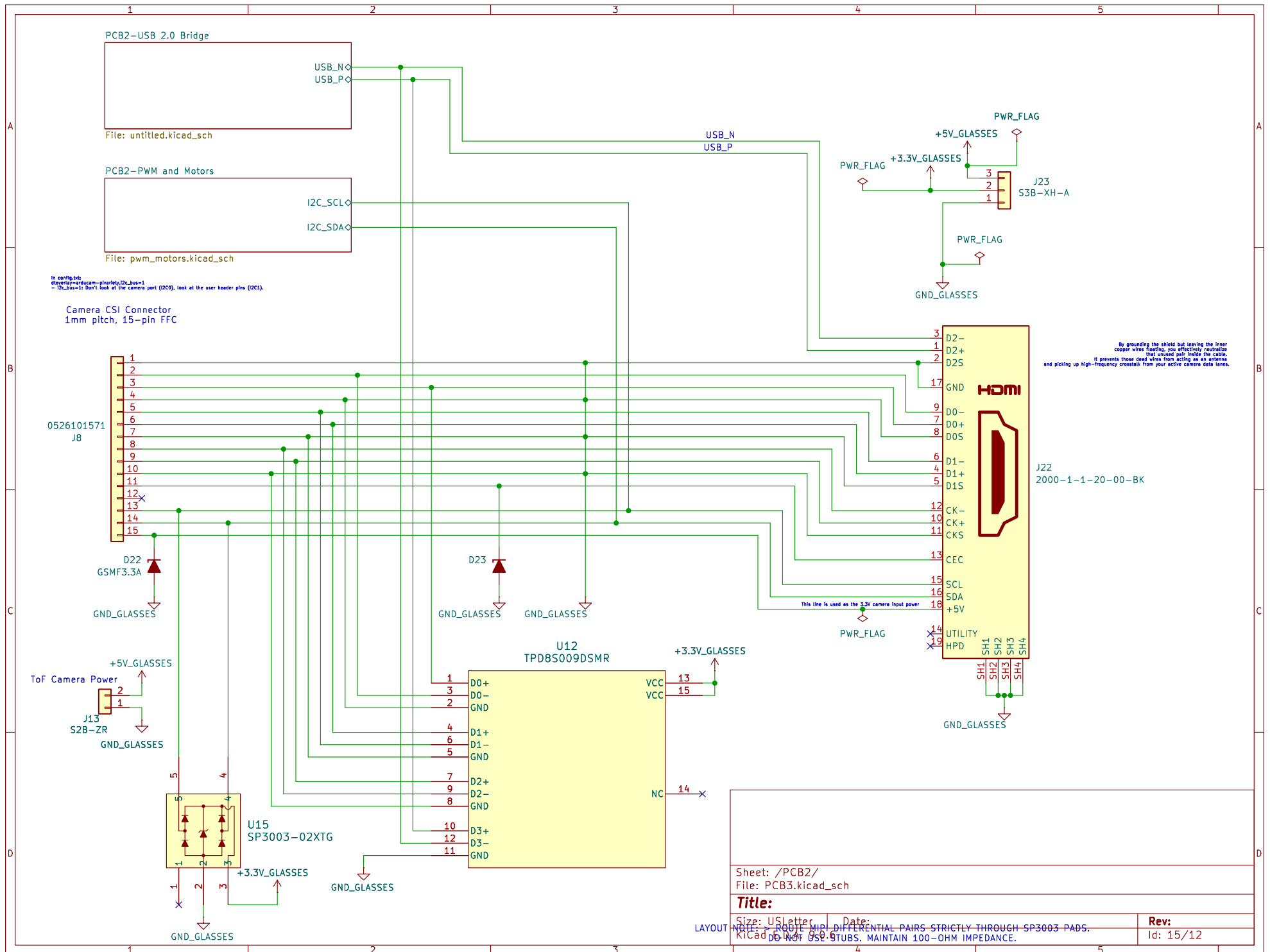
Sheet: /PCB1/PCB1-Board Connector A/
File: board_connector_a.kicad_sch

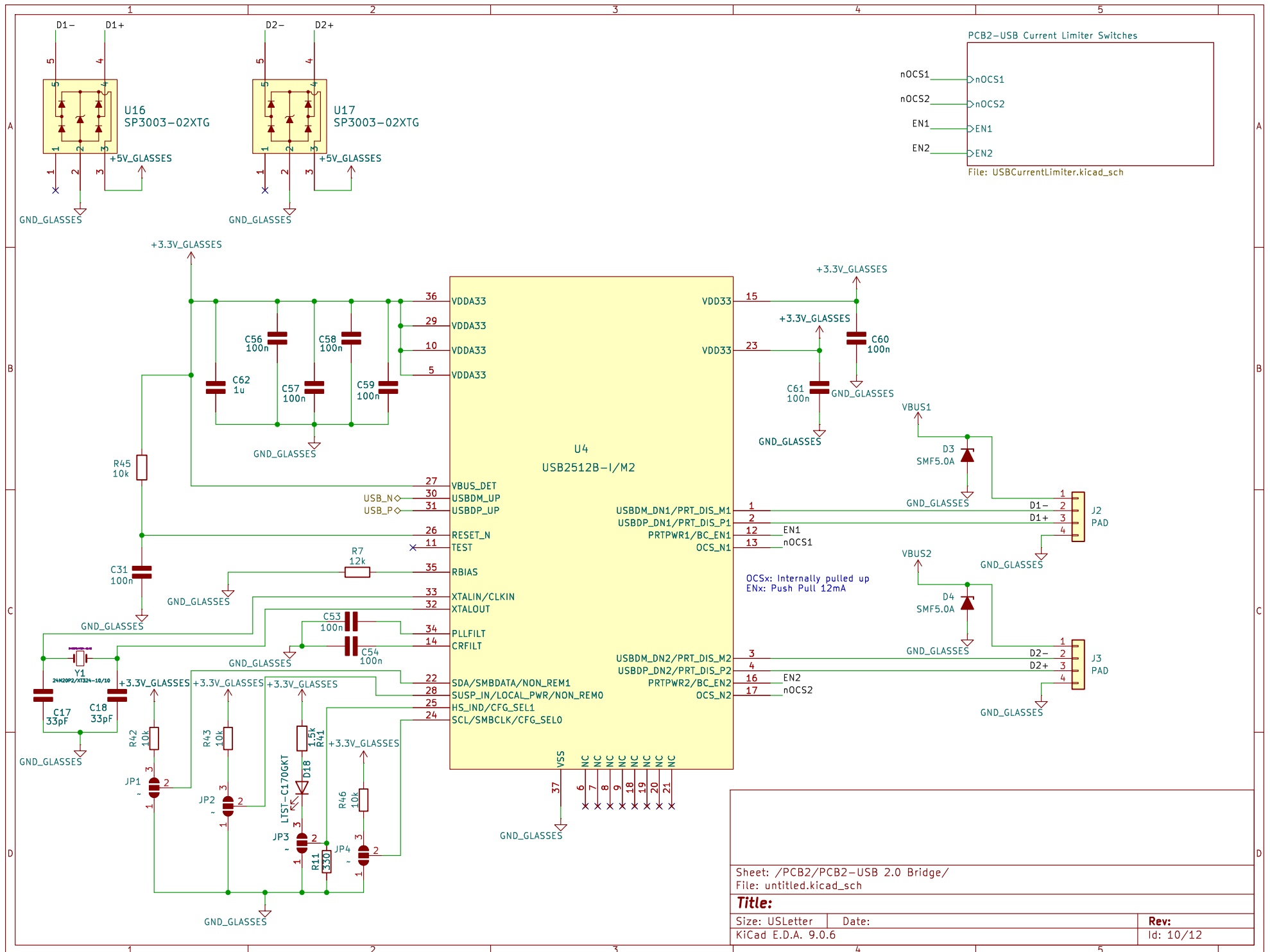
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Size: USLetter Date:
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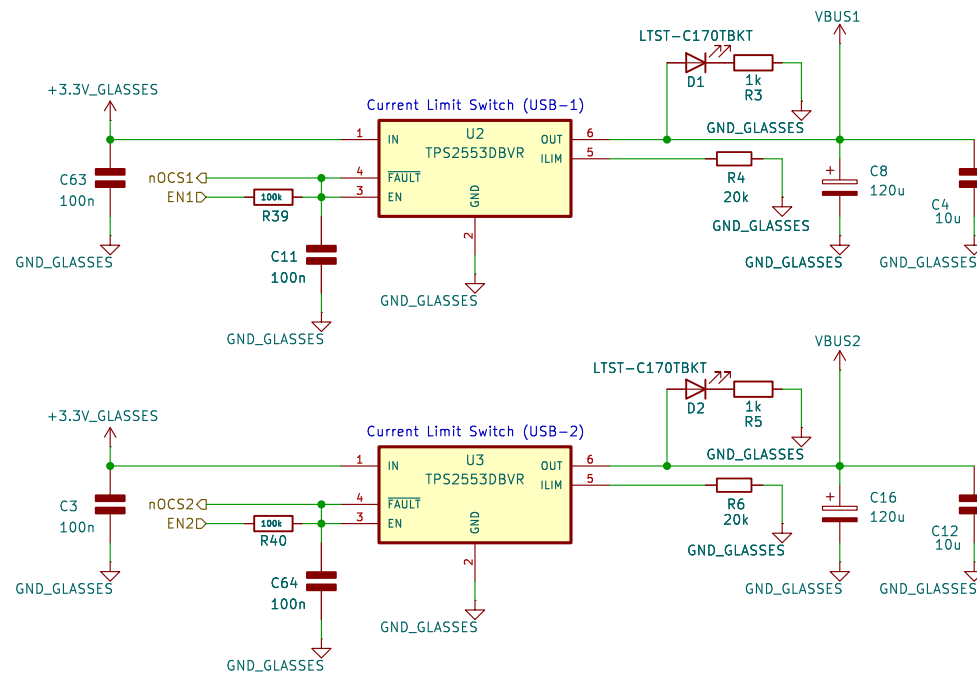
Rev:
Id: 18/12







auto-retry functionality and the ability to enable or disable with an external logic signal.
 Figure 28 shows how an external logic signal can drive EN through RFAULT and maintain auto-retry functionality.
 The resistor-capacitor time constant determines the auto-retry time-out period.



Sheet: /PCB2/PCB2-USB 2.0 Bridge/PCB2-USB Current Limiter Switches/
 File: USBCurrentLimiter.kicad_sch

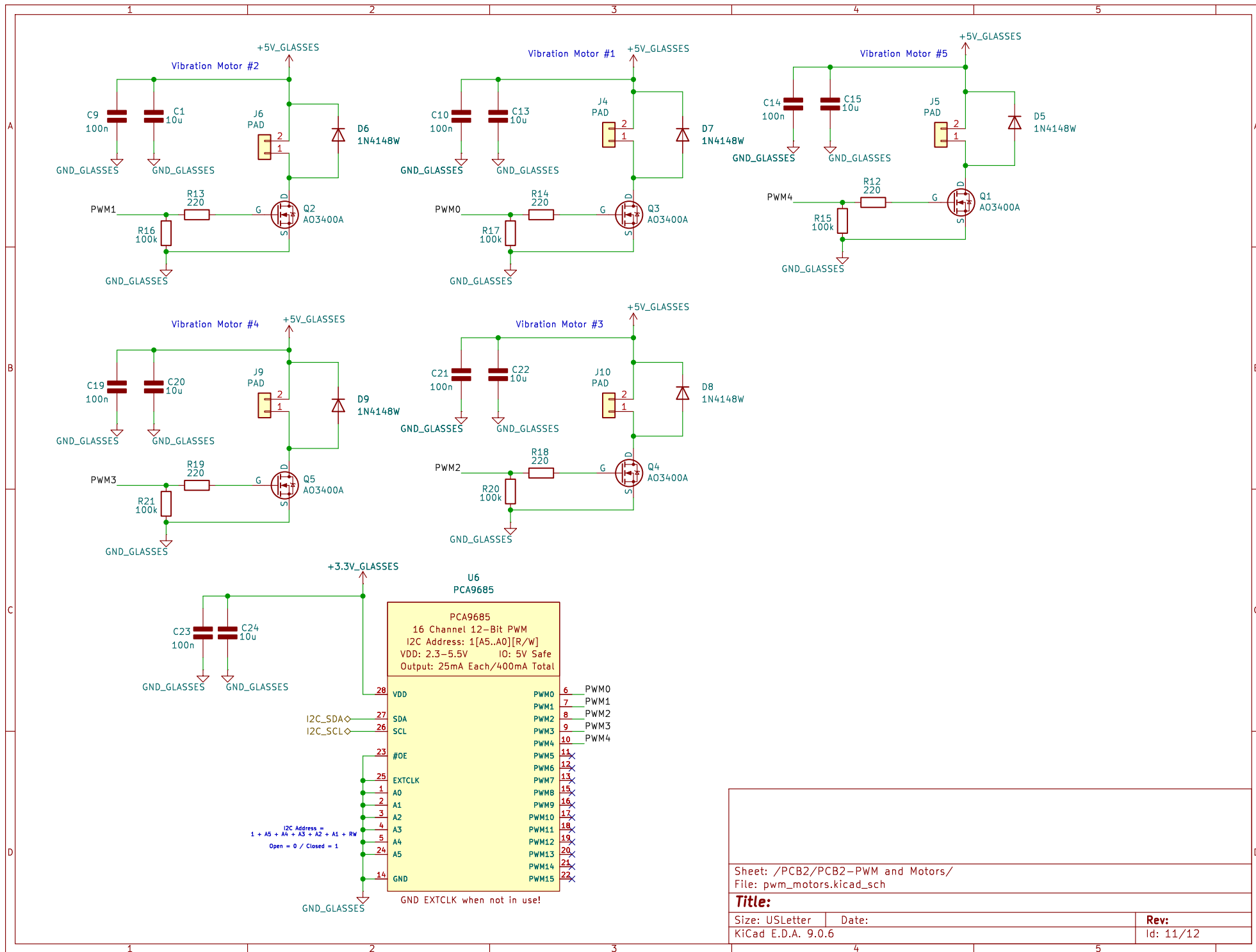
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Id: 11/12



Sheet: /PCB2/PCB2-PWM and Motors/
File: pwm_motors.kicad_sch

Title:

Size: USLetter Date:
KiCad E.D.A. 9.0.6

Rev:
Id: 11/12